

Make a VPC

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Purpose

This AWS lab allows me to learn more about networking and also using different platforms other than Cisco systems. In this lab, I learn how to create a VPC.

Lab Walkthrough (with Documentation)

Step 1: Create a VPC

VPC settings

Name tag - *optional*
Creates a tag with a key of 'Name' and a value that you specify.

IPv4 CIDR block [Info](#)

IPv6 CIDR block [Info](#)

☒ No IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

☐ IPv6 CIDR owned by me

Tenancy [Info](#)

Default ▼

Step 2: Make a Public Subnet

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

No preference ▼

IPv4 CIDR block [Info](#)

▼ **Tags - *optional***

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Step 3: Make a Private Subnet

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

No preference ▼

IPv4 CIDR block [Info](#)

🔍 10.0.2.0/24 ✕

▼ Tags - optional

Step 4: Make an Internet Gateway

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

Step 5: Attach Internet Gateway to the VPC

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

🔍 vpc-0f9167e028cfd9e6 ✕

▶ AWS Command Line Interface command

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Step 6: Make a Public Routing Table

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Name tag ⓘ

VPC* ⓘ

Step 7: Add a Route

Edit routes

Destination	Target	Status
10.0.0.0/16	local	active
0.0.0.0/0	igw-01fe3b25c08a54d6d	
Add route		

* Required

Cancel

Save routes

Step 8: Associate Route Table

Route table rtb-075c49f79933c23f9 (MyVPC Public Route Table)

Associated subnets

ⓘ

☐

Subnet ID

☒

subnet-0f8de159c12a06c6b | Public Sub...

10.0.1.0/24

☐

subnet-0857b86d96b9a22ea | Private S...

10.0.2.0/24

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Problems

In this lab, I had no problems or issues.

Lab Questions

1. **What does the acronym VPC stand for?** Virtual Private Cloud
2. **What is a VPC?** It is a secure, private cloud within a public cloud (virtual networks)
3. **What are your public and private subnet IDs?** My public subnet ID is subnet-0f8de159c12a06c6b and my private subnet ID is subnet-0857b86d96b9a22ea
4. **What is the purpose of your public subnet and the private subnet?** The public subnet has web servers and can send outbound traffic to the internet directly. The private subnet has the application and database servers and is able to access the internet using the NAT inside the public subnet.
5. **What are the two routes in the public subnet? (Look at route tables)** There is a route to 10.0.0.0/16 and 0.0.0.0/0
6. **What is the purpose of the internet gateway?** It allows your VPC to communicate with the internet
7. **Can resources launched in your private subnet communicate to the internet gateway directly?** No, it cannot communicate to the internet gateway directly like the public subnet can, however it can access the internet through a NAT gateway.
8. **What is a NAT? What is a NAT gateway?** NAT is network address translation. A NAT gateway is a very accessible AWS service that makes it easier to connect to the internet from within the private subnet in your VPC.
9. **What is the allowed block size for a VPC?** Between /16 and /28 netmask
10. **What is the minimum size for a VPC subnet?** The minimum is a /28 mask